Landing amplification for a vector state

Definition note for the two-state plant (Q8, 4 November 2026),
written before the statistic is used

1 Scalar statistic (reference)

For the scalar plant, $m_k = \hat{J}_x(x_k, u_k) - \hat{J}_u(x_k, u_k)K_k$ with $K_k = -\partial u_k/\partial x_k$ (direct gain). From the landing step k_ℓ (first k with $|x_k| \le 0.2$), $\Lambda = \max_{k_\ell \le k_1 \le k_2 < N} \sum_{k=k_1}^{k_2} \log|m_k|$, the largest contiguous sum, and $\Lambda^\star$ is the maximum over the 1000 training starts. Because $|\prod m_k| = \prod |m_k|$, this is the log of the largest gain of a contiguous stretch of the linearised closed loop through the surrogate.

2 Vector statistic

With $x_k \in \mathbb{R}^2$ and a scalar input, the surrogate Jacobians are $\hat{J}_x(x_k, u_k) \in \mathbb{R}^{2\times 2}$ and $\hat{J}_u(x_k, u_k) \in \mathbb{R}^2$, the policy's direct gain is the row vector $K_k = -\partial u_k/\partial x_k \in \mathbb{R}^{1\times 2}$ (at the current hidden state for a recurrent policy), and the closed-loop multiplier is the matrix

$$M_k = \hat{J}_x(x_k, u_k) - \hat{J}_u(x_k, u_k)\, K_k \in \mathbb{R}^{2\times 2}.$$

Landing step: $k_\ell = \min\{k : \|x_k\|_\infty \le 0.2\}$ ($k_\ell = N$ if none). For one start,

$$\Lambda = \max_{k_\ell \le k_1 \le k_2 < N} \log \left\| M_{k_2} M_{k_2-1} \cdots M_{k_1} \right\|_2, \qquad \Lambda \ge 0 \text{ by convention (empty stretch)},$$

with $\|\cdot\|_2$ the spectral norm, and $\Lambda^\star$ = max over starts. For $n_x = 1$ this is exactly the scalar statistic. Unlike the scalar case the log norm of a product is not the sum of the log norms (only $\log\|M_{k_2}\cdots M_{k_1}\| \le \sum \log\|M_k\|$), so the maximum is taken over all $O(N^2)$ stretches with running products; each running product is renormalised after every step and its log norm accumulated, which is exact.

3 What it is and is not

It is the largest finite-time growth factor of a deviation, over any contiguous stretch after landing, of the closed loop linearised along the surrogate rollout with the direct gain. It ignores the memory path of a recurrent policy, as the scalar direct-gain statistic does. A threshold for the two-state plant is not implied by the scalar threshold 12.1; if the statistic is used to predict traps, the threshold will be fixed and registered before the DPC runs it predicts.

4 Implementation

`src/plant2_core.py`, function `vector_lambda`: one `lax.scan` per start gives x_k and M_k; the running products for all k_1 are advanced together for $k_2 = 0, \ldots, N-1$ (only stretches with $k_1 \ge k_\ell$). Unit checks (tools/test_plant2.py): with diagonal M_k it equals the larger of the two scalar statistics; for a 1-D embedding it equals the scalar max-subarray value.